Predicting Monthly Salaries from Professional Profiles

This project uses the Koodiklinikka 2025 salary survey to compare models for predicting gross monthly salary from professional profiles.

Research question: How accurately can monthly salaries be predicted from professional profiles, and which modeling approach performs best?

The processed salary variable includes base salary and monthly commission.

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("koodiklinikka_2025.csv")

print("Dataset shape:", df.shape)
display(df.head())
```

```
Dataset shape: (682, 56)
```

	Timestamp	Palkansaaja vai laskuttaja	Siirtynyt palkansaaja/laskuttaja	Ikä	Sukupuoli	Suomen kielen taito
0	2026-02-09 08:08:48.457	Laskuttaja	Ei	36-40	mies	Yes
1	2026-02-09 08:10:51.389	Palkansaaja	Ei	46-50	mies	Yes
2	2026-02-09 08:13:13.891	Palkansaaja	Ei	36-40	mies	Yes
3	2026-02-09 08:14:20.316	Palkansaaja	Ei	26-30	mies	Yes
4	2026-02-09 08:14:32.538	Palkansaaja	Ei	41-45	mies	Yes

5 rows × 56 columns

1. Initial Data Inspection

The analysis focuses on employees rather than entrepreneurs or contractors. We first inspect missing values, salary values, and the categories of the candidate predictors before defining further cleaning rules.

In [2]:
```python
# Keep employees and preserve the original dataset
employees = df[
    df["Palkansaaja vai laskuttaja"] == "Palkansaaja"
].copy()

target = "Kuukausipalkka"

features = [
    "Työkokemus alalta (vuosina)",
    "Rooli (normalisoitu)",
    "Seniority",
    "Kaupunki",
    "Yrityksen koko",
    "Työaika",
]

print("All respondents:", len(df))
print("Employees:", len(employees))

print("\nMissing values:")
display(employees[[target] + features].isna().sum())

print("\nNumerical summary:")
display(
    employees[
        [target, "Työkokemus alalta (vuosina)", "Työaika"]
    ].describe()
)

print("\nNon-positive salary values:")
print((employees[target] <= 0).sum())

for column in [
    "Seniority",
    "Kaupunki",
    "Rooli (normalisoitu)",
    "Yrityksen koko",
]:
    print(f"\n{column}:")
    display(employees[column].value_counts(dropna=False))
```

```
All respondents: 682
Employees: 590

Missing values:
Kuukausipalkka                0
Työkokemus alalta (vuosina)   2
Rooli (normalisoitu)          3
Seniority                     8
Kaupunki                      8
Yrityksen koko                3
Työaika                       5
dtype: int64
Numerical summary:
```

	Kuukausipalkka	Työkokemus alalta (vuosina)	Työaika
count	590.000000	588.000000	585.000000
mean	6163.939294	12.736395	0.993750
std	4102.841160	7.073734	0.070580
min	0.000000	0.000000	0.600000
25%	4900.000000	7.000000	1.000000
50%	5800.000000	11.000000	1.000000
75%	6900.000000	17.250000	1.000000
max	92000.000000	42.000000	1.866667

```
Non-positive salary values:
2

Seniority:
Seniority
Senior                      299
Mid-level                   118
Staff / Principal / Lead    115
Manager or People Lead       42
Junior / Entry-level          8
NaN                           8
Name: count, dtype: int64
Kaupunki:
Kaupunki
PK-seutu        349
Tampere         110
Turku            50
Oulu             18
Jyväskylä        13
Etätyö           12
NaN               8
Kuopio            7
Joensuu           3
Ulkomaat          3
Vaasa             3
Lappeenranta      2
Hämeenlinna       2
Seinäjoki         1
US                1
Oslo              1
Dubai             1
Salo              1
Tallinna          1
Lahti             1
Copenhagen        1
Kokkola           1
Rovaniemi         1
Name: count, dtype: int64
Rooli (normalisoitu):
Rooli (normalisoitu)
```

```
Developer, Full-Stack
259
Architect, Software or Solutions
50
Developer, Backend
37
Developer, Frontend
22
Engineering Manager
20
Data Engineer
17
Developer, Mobile
16
Cloud Infrastructure Engineer
16
DevOps Engineer or Professional
15
Senior Executive (C-suite, VP, etc.)
11
Cybersecurity or InfoSec Professional
10
AI/ML Engineer
9
Product Manager
9
Developer, QA or Test
8
System Administrator
8
Founder, Technology or Otherwise
8
Project Manager
8
Developer, Embedded Applications or Devices
7
Developer, Desktop or Enterprise Applications
6
UX, Research Ops or UI Design Professional
5
Data Scientist
5
NaN
3
Developer, AI Apps or Physical AI
3
Data / Business Analyst
2
Infrastructure Engineer (On-Prem / Data Center / Hardware / Network)
2
Software Engineer
2
CTO
1
Chapter Lead
1
```

```
Business development director
1
Data Developer
1
AI-native do-it-all one-man-show projects
1
Senior Software Engineer
1
AI Specialist
1
Service desk
1
Sales
1
Platform Engineer
1
Produc Engineer
1
Developer
1
One person product team
1
Algorithm engineer
1
Developer Advocate
1
Integration Specialist
1
ICT Specialist. Tasks range from Linux servers to backend developmen
t to cloud solutions.      1
Recruiter
1
Educator
1
IT specialist
1
Test specialist
1
Business Analyst
1
Consultant
1
Integrator
1
Founding Engineer
1
Analytics Engineer
1
Team Lead
1
Development Manager
1
Tech lead
1
Support Engineer or Analyst
1
```

```
Startup CTO doing everything tech related
1
Applied Scientist
1
Name: count, dtype: int64
Yrityksen koko:
Yrityksen koko
11–50         115
101–500       105
1001–5000     104
501–1000       92
51–100         65
10000+         54
6–10           28
5001–10000     12
3–5            10
NaN             3
2               2
Name: count, dtype: int64
```

2. Sample Selection

The analysis includes employees with positive monthly salaries. Explicit foreign locations and missing locations are excluded.

Responses stating only "remote work" are retained as a separate location category. Given the Finnish community context, they are included under the assumption that they belong to the Finnish employment context. Their geographical location cannot be verified, which is a limitation.

Observations with missing predictor values are excluded using a complete-case approach. Extreme salary values are inspected separately.

In [3]:
```
data = employees.copy()
counts = {"Employees": len(data)}

# Standardize text values
text_cols = data.select_dtypes(include="object").columns
data[text_cols] = data[text_cols].apply(
    lambda s: s.str.strip().replace("", pd.NA)
)

# Require a positive salary
data = data[data[target].notna() & (data[target] > 0)].copy()
counts["Positive salary"] = len(data)

# Exclude explicit foreign locations; retain remote work
excluded = [
    "Ulkomaat", "US", "Dubai", "Oslo",
    "Tallinna", "Copenhagen",
]
data = data[
    data["Kaupunki"].notna() & ~data["Kaupunki"].isin(excluded)
].copy()
counts["Domestic or remote location"] = len(data)

# Require complete predictor values
data = data.dropna(subset=features).copy()
counts["Complete predictors"] = len(data)

display(pd.Series(counts, name="Observations").to_frame())

print("Highest reported monthly salaries:")
display(
    data.nlargest(5, target)[
        [target, "Rooli (normalisoitu)", "Seniority", "Työaika"]
    ]
)
```

	Observations
Employees	590
Positive salary	588
Domestic or remote location	572
Complete predictors	554

```
Highest reported monthly salaries:
```

	Kuukausipalkka	Rooli (normalisoitu)	Seniority	Työaika
87	92000.00	Developer, Full-Stack	Senior	1.000000
353	15800.00	Developer Advocate	Senior	1.000000
515	13200.00	Architect, Software or Solutions	Staff / Principal / Lead	1.066667
26	12063.33	Developer, Backend	Senior	1.000000
181	11800.00	Developer, Full-Stack	Staff / Principal / Lead	1.000000

3. Salary Distribution

One respondent reports a monthly salary of EUR 92,000, while the next-highest salary is EUR 15,800. The extreme value cannot be verified and is excluded from the primary analysis.

The sample including this observation is retained for sensitivity analysis. Other high salaries are not automatically excluded.

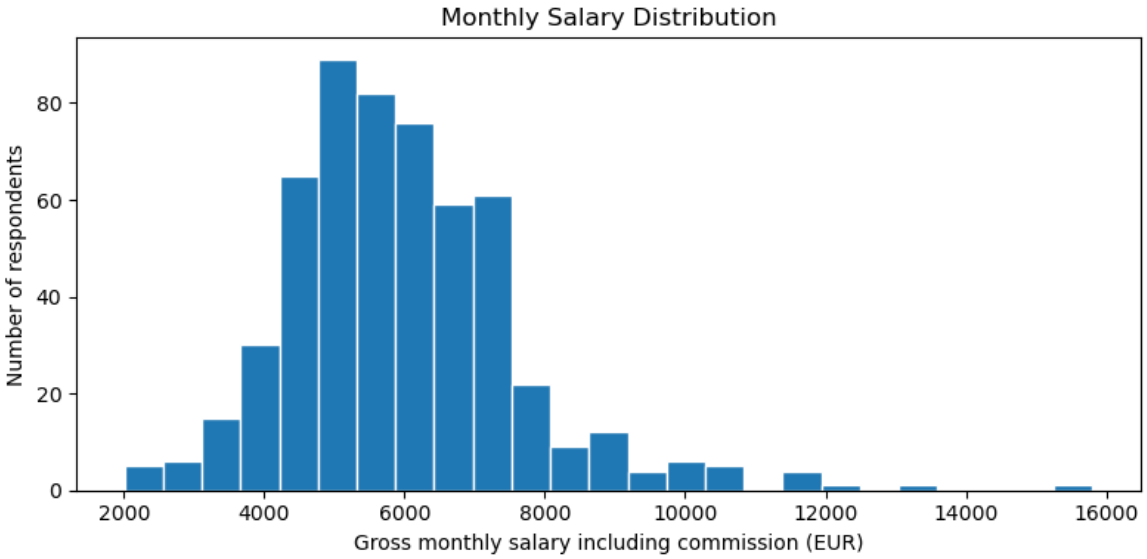
In [4]:
```
# Preserve the original sample for sensitivity analysis
data_with_extreme = data.copy()

# Exclude the single unverified extreme salary
data = data_with_extreme[
    data_with_extreme[target] != 92000
].copy()

print("Primary analysis sample:", len(data))
display(data[target].describe().round(2))

plt.figure(figsize=(8, 4))
plt.hist(data[target], bins=25, edgecolor="white")
plt.xlabel("Gross monthly salary including commission (EUR)")
plt.ylabel("Number of respondents")
plt.title("Monthly Salary Distribution")
plt.tight_layout()
plt.show()
```

```
Primary analysis sample: 553
count      553.00
mean      5954.76
std       1659.39
min       2009.00
25%       4900.00
50%       5800.00
75%       6800.00
max      15800.00
Name: Kuukausipalkka, dtype: float64
```



4. Salary and Professional Experience

Salary distributions are examined by industry experience and seniority. These descriptive comparisons help identify patterns and motivate the prediction models. They do not establish causal effects.

In [5]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

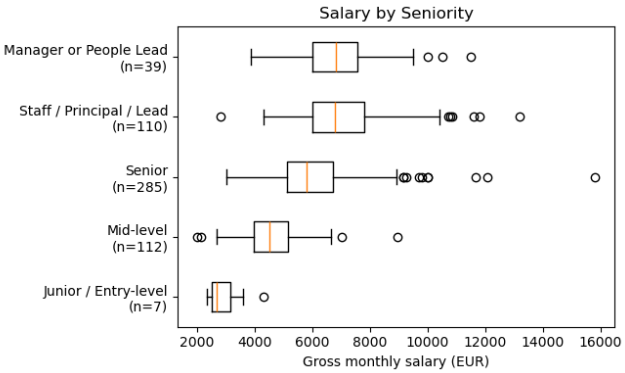
# Salary and years of experience
axes[0].scatter(
    data["Työkokemus alalta (vuosina)"],
    data[target],
    alpha=0.4,
)
axes[0].set_xlabel("Industry experience (years)")
axes[0].set_ylabel("Gross monthly salary (EUR)")
axes[0].set_title("Salary and Experience")

# Salary distribution by seniority
levels = [
    "Junior / Entry-level", "Mid-level", "Senior",
    "Staff / Principal / Lead", "Manager or People Lead",
]
values = [data.loc[data["Seniority"] == level, target] for level in
labels = [f"{level}\n(n={len(v)})" for level, v in zip(levels, valu

axes[1].boxplot(values, vert=False)
axes[1].set_yticks(range(1, len(levels) + 1))
axes[1].set_yticklabels(labels)
axes[1].set_xlabel("Gross monthly salary (EUR)")
axes[1].set_title("Salary by Seniority")

plt.tight_layout()
plt.show()
```





5. Training and Test Split

The data is randomly split into an 80% training set and a 20% test set. The training set will be used for preprocessing, cross-validation, and model selection. The test set will be reserved for final evaluation.

This evaluates prediction for held-out respondents from the same survey, not future salary development or representativeness of the entire sector.

In [6]:
```
from sklearn.model_selection import train_test_split

X = data[features].copy()
y = data[target].copy()

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
)

print("Training observations:", len(X_train))
print("Test observations:", len(X_test))
```

```
Training observations: 442
Test observations: 111
```

6. Preprocessing

Industry experience and working-time ratio are standardized. Categorical predictors are one-hot encoded, with categories appearing fewer than 10 times in each training fold grouped together.

Preprocessing is included in the modeling pipelines so that transformations are learned only from the training portion of each cross-validation fold.

In [7]:
```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline

numeric_features = [
    "Työkokemus alalta (vuosina)",
    "Työaika",
]

categorical_features = [
    "Rooli (normalisoitu)",
    "Seniority",
    "Kaupunki",
    "Yrityksen koko",
]

preprocessor = ColumnTransformer([
    ("numeric", StandardScaler(), numeric_features),
    ("categorical", OneHotEncoder(
        min_frequency=10,
        handle_unknown="infrequent_if_exist",
        drop="first",
        sparse_output=False,
    ), categorical_features),
])
```

7. Initial Model Comparison

A median-salary baseline is compared with Linear Regression, Ridge, and Random Forest using the same five cross-validation folds.

Mean absolute error (MAE), measured in euros per month, is the primary comparison metric. Lower values indicate better predictions. This is an initial comparison before limited hyperparameter tuning.

In [8]:
```
from sklearn.base import clone
from sklearn.dummy import DummyRegressor
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import KFold, cross_val_score

models = {
    "Median baseline": DummyRegressor(strategy="median"),
    "Linear Regression": LinearRegression(),
    "Ridge": Ridge(alpha=10),
    "Random Forest": RandomForestRegressor(
        n_estimators=200,
        max_depth=5,
        min_samples_leaf=5,
        random_state=42,
        n_jobs=1,
    ),
}

cv = KFold(n_splits=5, shuffle=True, random_state=42)
results = []

for name, model in models.items():
    pipeline = Pipeline([
        ("preprocessing", clone(preprocessor)),
        ("model", model),
    ])

    scores = -cross_val_score(
        pipeline, X_train, y_train,
        cv=cv,
        scoring="neg_mean_absolute_error",
        n_jobs=-1,
    )

    results.append({
        "Model": name,
        "CV MAE (EUR)": scores.mean(),
        "Fold SD (EUR)": scores.std(),
    })

comparison = pd.DataFrame(results).sort_values("CV MAE (EUR)")
display(comparison.round(2))
```

```
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_encoders.py:246: UserWarning: Found unknown categories in columns [0, 2] during transform. These unknown categories will be encoded as all zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_encoders.py:246: UserWarning: Found unknown categories in columns [0, 2] during transform. These unknown categories will be encoded as all zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_encoders.py:246: UserWarning: Found unknown categories in columns [0,
```

```
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
```

```
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
```

	Model	CV MAE (EUR)	Fold SD (EUR)
3	Random Forest	901.95	56.53
2	Ridge	903.46	43.66
1	Linear Regression	909.26	46.86
0	Median baseline	1235.89	146.21

8. Limited Hyperparameter Tuning

Ridge and Random Forest are tuned using the same five-fold cross-validation scheme and MAE criterion. The search is deliberately small because the dataset is limited.

Cross-validation scores guide model selection; the test set remains unused until the final evaluation.

In [9]:
```
from sklearn.model_selection import GridSearchCV

searches = {}
tuning_results = []

settings = {
    "Ridge": (
        Ridge(),
        {"model__alpha": [0.1, 1, 10, 100]},
    ),
    "Random Forest": (
        RandomForestRegressor(
            n_estimators=200, random_state=42, n_jobs=1
        ),
        {
            "model__max_depth": [3, 5, None],
            "model__min_samples_leaf": [3, 5, 10],
        },
    ),
}

for name, (model, parameters) in settings.items():
    pipeline = Pipeline([
        ("preprocessing", clone(preprocessor)),
        ("model", model),
    ])

    search = GridSearchCV(
        pipeline, parameters,
        scoring="neg_mean_absolute_error",
        cv=cv,
        n_jobs=-1,
    )
    search.fit(X_train, y_train)
    searches[name] = search

    tuning_results.append({
        "Model": name,
        "Best CV MAE (EUR)": -search.best_score_,
        "Parameters": search.best_params_,
    })

display(pd.DataFrame(tuning_results).round(2))
```

```
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_encoders.py:246: UserWarning: Found unknown categories in columns [0, 2] during transform. These unknown categories will be encoded as all zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_encoders.py:246: UserWarning: Found unknown categories in columns [0, 2] during transform. These unknown categories will be encoded as all zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_encoders.py:246: UserWarning: Found unknown categories in columns [0,
```

```
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
```

```
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
```

```
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
```

```
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
```

```
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
```

```
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
3] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0,
2] during transform. These unknown categories will be encoded as all
zeros
  warnings.warn(
```

	Model	Best CV MAE (EUR)	Parameters
0	Ridge	902.21	{'model__alpha': 1}
1	Random Forest	894.78	{'model__max_depth': 5, 'model__min_samples_le...

9. Final Test Evaluation

Random Forest is selected based on the lowest cross-validation MAE. Its advantage over Ridge is small.

The selected model is evaluated on the held-out test set and compared with the training-set median baseline. Test results are not used for further model selection or hyperparameter tuning.

MAE and RMSE are reported in euros per month. R^2 measures improvement in squared error relative to predicting the test-set mean.

In [10]:
```
from sklearn.metrics import mean_absolute_error, mean_squared_error

# GridSearchCV has already refitted this pipeline on all training d
final_model = searches["Random Forest"].best_estimator_

print("Selected model: Random Forest")
print("Parameters:", searches["Random Forest"].best_params_)

y_pred = final_model.predict(X_test)

baseline = DummyRegressor(strategy="median")
baseline.fit(X_train, y_train)
baseline_pred = baseline.predict(X_test)

test_results = []

for name, predictions in [
    ("Median baseline", baseline_pred),
    ("Selected Random Forest", y_pred),
]:
    test_results.append({
        "Model": name,
        "MAE (EUR)": mean_absolute_error(y_test, predictions),
        "RMSE (EUR)": mean_squared_error(y_test, predictions) ** 0.
        "R²": r2_score(y_test, predictions),
    })

display(pd.DataFrame(test_results).round(2))
```

```
Selected model: Random Forest
Parameters: {'model__max_depth': 5, 'model__min_samples_leaf': 3}
```

```
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0]
during transform. These unknown categories will be encoded as all ze
ros
  warnings.warn(
```

	Model	MAE (EUR)	RMSE (EUR)	R²
0	Median baseline	1156.73	1614.01	-0.01
1	Selected Random Forest	845.85	1253.32	0.39

10. Prediction Errors

Test predictions are compared with reported salaries. The diagonal line represents a perfect prediction.

Residuals are defined as reported salary minus predicted salary. Positive residuals indicate underprediction; negative residuals indicate overprediction. These plots are used for interpretation, not further tuning.

In [11]:
```
residuals = y_test - y_pred

fig, axes = plt.subplots(1, 2, figsize=(11, 4))

# Actual versus predicted salaries
axes[0].scatter(y_test, y_pred, alpha=0.6)
limits = [
    min(y_test.min(), y_pred.min()),
    max(y_test.max(), y_pred.max()),
]
axes[0].plot(limits, limits, "--", color="black")
axes[0].set_xlabel("Reported monthly salary (EUR)")
axes[0].set_ylabel("Predicted monthly salary (EUR)")
axes[0].set_title("Reported vs. Predicted Salary")

# Residuals versus predicted salaries
axes[1].scatter(y_pred, residuals, alpha=0.6)
axes[1].axhline(0, linestyle="--", color="black")
axes[1].set_xlabel("Predicted monthly salary (EUR)")
axes[1].set_ylabel("Reported minus predicted salary (EUR)")
axes[1].set_title("Prediction Residuals")

plt.tight_layout()
plt.show()
```



Interpretation of Test Errors

The model improves on the median baseline, but its predictions cover a narrower range than the reported salaries. Several high salaries are substantially underpredicted.

The test MAE is approximately EUR 846 per month, while the RMSE is approximately EUR 1,253. The larger RMSE reflects the influence of some substantial prediction errors.

These results suggest that the model captures broad salary patterns but does not reliably identify unusually high compensation. It should therefore not be treated as a precise salary recommendation for an individual employee.

11. Sensitivity to the Extreme Salary Observation

The excluded EUR 92,000 salary observation is added to the training set. The selected model settings and the original test set remain unchanged.

This analysis examines sensitivity to including the observation during training. It does not evaluate prediction accuracy for extreme salaries or guide further model tuning.

In [12]:
```
extreme = data_with_extreme[
    data_with_extreme[target] == 92000
]

X_train_extreme = pd.concat([X_train, extreme[features]])
y_train_extreme = pd.concat([y_train, extreme[target]])

sensitivity_model = clone(final_model)
sensitivity_model.fit(X_train_extreme, y_train_extreme)
sensitivity_pred = sensitivity_model.predict(X_test)

sensitivity_results = []

for name, predictions in [
    ("Primary analysis", y_pred),
    ("Extreme observation added to training", sensitivity_pred),
]:
    sensitivity_results.append({
        "Scenario": name,
        "Test MAE (EUR)": mean_absolute_error(y_test, predictions),
        "Test RMSE (EUR)": mean_squared_error(y_test, predictions) 
        "Test R²": r2_score(y_test, predictions),
    })

print("Extreme observations added:", len(extreme))
display(pd.DataFrame(sensitivity_results).round(2))
```

```
Extreme observations added: 1
```

```
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_encoders.py:246: UserWarning: Found unknown categories in columns [0] during transform. These unknown categories will be encoded as all zeros
  warnings.warn(
```

	Scenario	Test MAE (EUR)	Test RMSE (EUR)	Test R²
0	Primary analysis	845.85	1253.32	0.39
1	Extreme observation added to training	1187.48	2336.54	-1.12

Sensitivity Analysis Interpretation

Adding the EUR 92,000 observation to the training set increased test MAE from EUR 845.85 to EUR 1,187.48 and RMSE from EUR 1,253.32 to EUR 2,336.54. Test R^2 decreased from 0.39 to -1.12.

The selected model is therefore sensitive to this extreme training observation. This does not establish that the reported salary is incorrect. Its exclusion from the primary analysis remains a documented data-quality decision, and the results are conditional on that decision.

The original test set and model settings were unchanged.

12. Sensitivity to Remote-Only Location Responses

Respondents reporting only "remote work" are removed from the training set. Model settings remain unchanged.

Both the primary model and the sensitivity model are evaluated on the same test subset with explicitly reported Finnish locations. This isolates the effect of retaining remote-only respondents during training.

The analysis does not establish the geographical location of remote respondents and is not used for further model tuning.

In [13]:
```
# Exclude remote-only locations from training
domestic_train = X_train["Kaupunki"] != "Etätyö"
domestic_test = X_test["Kaupunki"] != "Etätyö"

location_model = clone(final_model)
location_model.fit(
    X_train.loc[domestic_train],
    y_train.loc[domestic_train],
)

# Compare both models on the same domestic test subset
X_domestic = X_test.loc[domestic_test]
y_domestic = y_test.loc[domestic_test]

location_results = []

for name, model in [
    ("Primary model", final_model),
    ("Remote-only training responses excluded", location_model),
]:
    predictions = model.predict(X_domestic)

    location_results.append({
        "Scenario": name,
        "MAE (EUR)": mean_absolute_error(y_domestic, predictions),
        "RMSE (EUR)": mean_squared_error(y_domestic, predictions) *
        "R²": r2_score(y_domestic, predictions),
    })

print("Remote-only training responses removed:", (~domestic_train).
print("Shared domestic test observations:", len(y_domestic))
display(pd.DataFrame(location_results).round(2))
```

```
Remote-only training responses removed: 10
Shared domestic test observations: 109
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0]
during transform. These unknown categories will be encoded as all ze
ros
  warnings.warn(
/opt/software/lib/python3.10/site-packages/sklearn/preprocessing/_en
coders.py:246: UserWarning: Found unknown categories in columns [0]
during transform. These unknown categories will be encoded as all ze
ros
  warnings.warn(
```

	Scenario	MAE (EUR)	RMSE (EUR)	R²
0	Primary model	855.30	1263.66	0.39
1	Remote-only training responses excluded	879.74	1297.51	0.35

In [ ]: